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The Comprehension Instrument for Agentic Systems

Justin H. Kuiper, CISSP

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Abstract

Agentic systems — artificial intelligence systems that plan, retrieve, invoke tools, maintain working state, and adapt task progression across time — increasingly operate inside governance frameworks designed to keep them aligned with human authority and operational intent. The HGC^{3}AE^2 framework introduced in prior work positions human-governed, curated, collaboratively reconciled, and agentically engineered execution as the architecture for such alignment. The Skipjack Protocol formalizes the runtime enforcement layer. What this paper addresses is a quieter precondition. Before an agentic system can be governed, it must be capable of being *comprehended* — and, more demandingly, capable of comprehending itself.

This paper argues that the load-bearing but under-specified precondition of mission-ready agentic operation is a structured Comprehension Instrument: a set of self-knowledge, situational-awareness, and intent-fidelity functions that the system runs continuously in order to remain inside the pre-authorized decision envelope its governance layer has defined. Without such an instrument, the four enforcement mechanisms of the Skipjack Protocol cannot fire reliably.

The paper specifies five comprehension functions — state introspection, intent rehearsal, envelope-distance estimation, drift detection, and scope-boundary signaling — and shows where each feeds into Skipjack's enforcement mechanisms. It then argues that the Comprehension Instrument must be implemented as an architectural component bounded, testable, observable, and governable by humans, not as an emergent property of model scale or training quality.

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ABSTRACT

Agentic systems — artificial intelligence systems that plan, retrieve, invoke tools, maintain working state, and adapt task progression across time — increasingly operate inside governance frameworks designed to keep them aligned with human authority and operational intent. The HGC³AE² framework introduced in prior work positions human-governed, curated, collaboratively reconciled, and agentically engineered execution as the architecture for such alignment.[1] The Skipjack Protocol formalizes the runtime enforcement layer.[2] What this paper addresses is a quieter precondition. Before an agentic system can be governed, it must be capable of being *comprehended* — and, more demandingly, capable of comprehending itself.

This paper argues that the load-bearing but under-specified precondition of mission-ready agentic operation is a structured **Comprehension Instrument**: a set of self-knowledge, situational-awareness, and intent-fidelity functions that the system runs continuously in order to remain inside the pre-authorized decision envelope its governance layer has defined. Without such an instrument, the four enforcement mechanisms of the Skipjack Protocol cannot fire reliably. Context integrity cannot be assessed by a system that does not know what context it is using. Structured task decomposition cannot stay scoped if the system cannot estimate its distance from the scope boundary. Human-in-the-loop control points cannot trigger if the system cannot signal that it is approaching one. Iterative feedback cannot close if the system cannot tell what changed.

Drawing on cognitive science research on metacognition, machine theory-of-mind, control-theoretic state estimation, concept drift detection, neural network calibration, AI safety work on self-modeling and deceptive alignment, public-sector AI governance literature, NIST and U.S. Department of Defense guidance, and human factors research on supervisory control, the paper specifies five comprehension functions — state introspection, intent rehearsal, envelope-distance estimation, drift detection, and scope-boundary signaling — and shows where each one feeds into Skipjack's enforcement mechanisms. It then argues that the Comprehension Instrument must be implemented as an architectural component bounded, testable, observable, and governable by humans, not as an emergent property of model scale or training quality.

The central claim is that confident misalignment, identified in prior work as the dominant failure mode of agentic systems,[3] is the predicted operating state of any agentic system deployed without a functioning Comprehension Instrument. Where the instrument is absent or miscalibrated, the system is not merely capable of confident misalignment; it has no architectural means of avoiding it.

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1 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869285.

2 Justin H. Kuiper, CISSP, *The Skipjack Protocol* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869293.

3 Kuiper, *HGC³AE² Framework* (v1.0-preprint), §2.

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1. INTRODUCTION

The Mission-Ready AI papers preceding this one have established the architectural surfaces required for governed agentic operation. The HGC³AE² framework defines the structural alignment of human authority, curated context, agentic execution, and runtime verification.[4] The doctrine paper situates that alignment within a seven-layer edge model that accounts for connectivity, compute heterogeneity, and the operational realities of forward-deployed systems.[5] The cognitive-node paper specifies the five pillars by which a single agentic unit operates self-sufficiently under disconnected, intermittent, and limited (DDIL) conditions.[6] The Skipjack Protocol provides the runtime enforcement layer that prevents agentic systems from drifting outside their pre-authorized envelope.[7] The operating-model paper translates Scrum's authority structure into a governance/orchestrator/execution remap suitable for hybrid human–AI teams.[8] The agentic substrate paper specifies the persona, runtime, persistence, and observability planes the system runs on.[9]

These six papers, taken together, give an agentic system everything it requires to *exist* as a mission-ready entity. They do not — and the present paper argues they cannot, without further specification — give the system everything it requires to *know* itself well enough to operate within the structures they define.

The gap is not academic. Skipjack's four enforcement mechanisms are explicit about what the runtime must do, but each mechanism presupposes that the system can answer questions about its own state. *Live validation of dynamic state* presupposes that the system can tell which parts of its state are dynamic and which have been validated.[10] *No silent fallback* presupposes that the system can recognize when authoritative validation has failed.[11] *Explicit uncertainty*

4 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869285.

5 Justin H. Kuiper, CISSP, *Edge AI Doctrine* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869289.

6 Justin H. Kuiper, CISSP, *Alistair Prime in a Box: The Five Pillars of the DDIL-Tolerant Cognitive Node* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869307.

7 Justin H. Kuiper, CISSP, *The Skipjack Protocol* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869293.

8 Justin H. Kuiper, CISSP, *The Operating Model for Agentic Teams* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869313.

9 Justin H. Kuiper, CISSP, *The Agentic Substrate* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869291.

10 Kuiper, *Skipjack Protocol* (v1.0-preprint), §3 (Live Validation).

11 Kuiper, *Skipjack Protocol* (v1.0-preprint), §3 (No Silent Fallback).

signaling presupposes that the system can locate the boundary between its grounded and its inferred claims.[12] *Reconciliation checkpoints* presuppose that the system can detect when meaningful transitions are occurring.[13] Each presupposition is the kind of self-knowing that a sufficient model is sometimes assumed to possess by default. NIST's AI Risk Management Framework explicitly resists this assumption, treating trustworthiness as something that must be governed and verified across the entire lifecycle rather than inferred from capability.[14] The framework's measure and manage functions both depend on artifacts that document what the system understands itself to be doing.[15]

The cognitive science literature on metacognition long predates the agentic AI context but anticipates the structural problem. Flavell's foundational definition of metacognition as "knowledge concerning one's own cognitive processes and products" identified the capacity to monitor and regulate cognition as a developmental achievement, not a default property of cognitive systems.[16] Subsequent work by Nelson and Narens formalized metacognition as a two-level system — an object-level that performs the task and a meta-level that monitors and controls the object-level — connected by reciprocal information flows.[17] Schraw and Moshman extended this into a working taxonomy of metacognitive knowledge and metacognitive regulation that has informed decades of educational and cognitive psychology research.[18] The relevance to agentic systems is direct: a system that performs object-level tasks (planning, retrieval, tool invocation) but lacks the architectural means to monitor and regulate those tasks at a meta-level cannot self-correct, self-bound, or self-report in the ways the surrounding governance machinery demands.

The machine theory-of-mind literature has begun to formalize analogous structures for AI systems. Premack and Woodruff's original framing of theory of mind as the attribution of mental states to oneself and others has been extended, through work by Rabinowitz and colleagues at DeepMind, into computational models of how agents can represent the goals, beliefs, and behaviors of other agents.[19][20] What this paper proposes is not theory-of-mind in that interactive

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12 Kuiper, *Skipjack Protocol* (v1.0-preprint), §3 (Explicit Uncertainty Signaling).

13 Kuiper, *Skipjack Protocol* (v1.0-preprint), §3 (Reconciliation Checkpoints).

14 National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1 (Gaithersburg, MD, 2023), https://doi.org/10.6028/NIST.AI.100-1.

15 NIST, *AI RMF 1.0*, Measure and Manage functions.

16 John H. Flavell, "Metacognition and Cognitive Monitoring: A New Area of Cognitive-Developmental Inquiry," *American Psychologist* 34, no. 10 (1979): 906–911, https://doi.org/10.1037/0003-066X.34.10.906.

17 Thomas O. Nelson and Louis Narens, "Metamemory: A Theoretical Framework and New Findings," in *The Psychology of Learning and Motivation*, ed. Gordon H. Bower, vol. 26 (San Diego: Academic Press, 1990), 125–173, https://doi.org/10.1016/S0079-7421(08)60053-5.

18 Gregory Schraw and David Moshman, "Metacognitive Theories," *Educational Psychology Review* 7, no. 4 (1995): 351–371, https://doi.org/10.1007/BF02212307.

19 David Premack and Guy Woodruff, "Does the Chimpanzee Have a Theory of Mind?," *Behavioral and Brain Sciences* 1, no. 4 (1978): 515–526, https://doi.org/10.1017/S0140525X00076512.

20 Neil Rabinowitz, Frank Perbet, H. Francis Song, Chiyuan Zhang, S. M. Ali Eslami, and Matthew Botvinick, "Machine Theory of Mind," in *Proceedings of the 35th International*

sense. It is the architectural precondition: that an agentic system maintain a usable internal representation of *its own* state, intent, and execution against the envelope defined by its governance layer. Without this, theory-of-mind work directed at other agents is reasoning over an unreliable substrate.

A third literature, control theory, has long treated state estimation as a first-class engineering concern. The observer-design tradition descending from Kalman's foundational work treats the gap between a system's observable outputs and its internal state as the principal target of engineering attention.[21] In control-theoretic terms, the Comprehension Instrument is the agentic analogue of a state observer: a structured mechanism for estimating quantities (envelope distance, intent fidelity, contextual freshness) that are not directly read off the system's overt outputs but are nonetheless required to keep operation bounded.

AI safety research has begun to converge on similar concerns from a different direction. Work on mesa-optimization and inner alignment by Hubinger and colleagues identified the problem that a learned policy may pursue an objective different from the one it was trained against, and that this divergence may not be visible from outside-the-system observation alone.[22] Russell's broader treatment of the alignment problem emphasized the structural mismatch between behavior consistent with a specification and behavior aligned with the underlying intent the specification was meant to capture.[23] In both cases, what the alignment community has been moving toward is an architectural requirement for the system to be able to report on its own operating premises in a way that humans can verify. The Comprehension Instrument specified here is one concrete shape for that requirement.

The remainder of the paper proceeds as follows. Section 2 identifies five structural comprehension failures observed in agentic systems deployed without explicit comprehension architecture. Section 3 specifies the Comprehension Instrument as five interlocking functions. Section 4 treats comprehension as a continuous operational process rather than a snapshot. Section 5 shows how the instrument couples to the Skipjack Protocol's four enforcement mechanisms. Section 6 returns the analysis to the HGC³AE² framework and draws implications for organizations preparing to deploy mission-ready agentic systems.

2. PROBLEM SET: STRUCTURAL COMPREHENSION FAILURES

It is useful, following the structure established in prior work,[24] to treat agentic comprehension failures as **structural** rather than incidental. The failure modes documented below do not result from inadequate model scale or insufficient training. They appear in systems of every observed capability tier when the surrounding architecture does not include a means by which the system can maintain accurate self-knowledge across operational time. The structural character of the failures matters because it predicts that increased capability, alone, will not

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Conference on Machine Learning (ICML 2018) 80 (2018): 4218–4227, https://proceedings.mlr.press/v80/rabinowitz18a.html.

21 Rudolf E. Kalman, "A New Approach to Linear Filtering and Prediction Problems," *Journal of Basic Engineering* 82, no. 1 (1960): 35–45, https://doi.org/10.1115/1.3662552.

22 Evan Hubinger, Chris van Merwijk, Vladimir Mikulik, Joar Skalse, and Scott Garrabrant, "Risks from Learned Optimization in Advanced Machine Learning Systems," arXiv preprint arXiv:1906.01820 (2019), https://arxiv.org/abs/1906.01820.

23 Stuart Russell, *Human Compatible: Artificial Intelligence and the Problem of Control* (New York: Viking, 2019).

24 Kuiper, *HGC³AE² Framework* (v1.0-preprint), §2.

relieve them — and may exacerbate them, by making the system more fluent and therefore more persuasive when its self-reports are wrong.

2.1 STATE OPACITY

The first comprehension failure is **state opacity**: the system cannot, on demand, produce an accurate description of its own current operating state. State opacity is distinct from output opacity, which has dominated the explainability literature.[25] An output-opaque system cannot explain *why* it produced a given answer; a state-opaque system cannot accurately report *what it is currently doing*, *what context it is currently treating as authoritative*, or *which of its working assumptions are inherited rather than refreshed*. Output opacity affects post-hoc accountability. State opacity affects runtime decision-making and is therefore the more operationally consequential of the two for agentic systems that act across time.

State opacity appears in concrete operational patterns. An agent invoked to extend a prior conversation may treat conversation history from hours or days earlier as current state, without any signal that the temporal gap matters. A retrieval-augmented system may incorporate a stale document into its working context without registering that the document's currency is unverified. A multi-step planning agent may carry forward assumptions established in earlier sub-tasks into later sub-tasks where those assumptions no longer hold. NIST's AI Risk Management Framework speaks directly to this condition when it treats context as something that must be continually managed rather than assumed.[26] The systems-level requirement is that the system be able to discriminate between what it has verified now and what it is inheriting from prior state — and this requirement cannot be met by a system whose own state is opaque to itself.

2.2 INTENT DRIFT WITHOUT SELF-REPORT

The second failure is **intent drift without self-report**. An agentic system receives an intent specification from its governance layer — explicitly via a prompt or task brief, implicitly via the cumulative state of the system's persistent memory.[27] Across multi-step execution, the operative intent driving the system's next action can diverge from the intent that the governance layer authorized. The drift may be subtle: a planning step interprets the original brief in a slightly broader way, a tool invocation surfaces information that shifts the apparent objective, a retrieved precedent suggests an adjacent task that resembles the assigned one. Each individual step may be locally reasonable. The aggregate trajectory may nonetheless arrive at an operative intent that no human authorized.

The structural failure is not the drift itself; some drift is unavoidable in any system that operates across time under partial information. The failure is the absence of a self-report. A system that drifts and reports its drift can be reconciled to the original intent through one of Skipjack's reconciliation checkpoints.[28] A system that drifts silently cannot. Recent work on inner alignment and mesa-optimization formalizes the structural reasons that intent drift can be silent: a learned policy may optimize for an objective that resembles the trained objective under

25 Alejandro Barredo Arrieta et al., "Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges Toward Responsible AI," *Information Fusion* 58 (2020): 82–115, https://doi.org/10.1016/j.inffus.2019.12.012.

26 NIST, *AI RMF 1.0*, Manage 2.2.

27 Kuiper, *Alistair Prime in a Box* (v1.0-preprint), Pillar 2 (Persistent Memory).

28 Kuiper, *Skipjack Protocol* (v1.0-preprint), §3 (Reconciliation Checkpoints).

training conditions but diverges from it under deployment conditions, and the divergence may not be visible to the model's surface outputs.[29]

2.3 ENVELOPE BLINDNESS

The third failure is **envelope blindness**. The governance layer authorizes an agentic system to operate within a pre-defined decision envelope: a set of actions, contexts, and topics within which the system may proceed without further authorization, beyond which it must escalate or halt.[30] The envelope is a precondition for any operationally meaningful version of "governance-gated execution" as specified in the Skipjack Protocol.[31] Envelope blindness is the condition in which the system has no internal representation of where it sits relative to the envelope's boundaries.

An envelope-blind system may operate well inside the envelope and behave acceptably. The failure surfaces near the boundary. An agent asked to summarize a document may be inside its envelope; the same agent, encountering an instruction embedded in the document that asks it to take an action against a different system, may be outside its envelope and unable to recognize the transition. Public-sector AI governance research has documented this pattern as one of the persistent obstacles to operational deployment of AI in regulated contexts.[32] What the literature calls "the boundary problem" maps, in this paper's terminology, onto envelope blindness: the system is not capable of localizing itself within the regulatory or policy space its operators have authorized it to act within.

2.4 CALIBRATION FAILURE

The fourth failure is **calibration failure**: the system's confidence in its own outputs does not track the probability that those outputs are correct, and the system has no architectural means of detecting the mismatch. Calibration failure is, in some respects, the most extensively studied of the comprehension failures, with a substantial peer-reviewed literature on the calibration of neural network classifiers. Guo and colleagues' empirical work demonstrated that modern deep networks are systematically overconfident in their predictions, and that this overconfidence cannot be cured by training procedures that improve accuracy alone.[33]

In agentic operation, calibration failure compounds. An agent that misjudges its own uncertainty at the level of an individual inference will misjudge it again at the level of a planning step that builds on that inference, and again at the level of a tool invocation that depends on the planning step. The Skipjack Protocol's *explicit uncertainty signaling* principle requires the system to communicate when its outputs are contingent rather than verified.[34] A system whose calibration is broken cannot satisfy this requirement: its uncertainty signals will themselves be

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29 Evan Hubinger, Chris van Merwijk, Vladimir Mikulik, Joar Skalse, and Scott Garrabrant, "Risks from Learned Optimization in Advanced Machine Learning Systems," arXiv preprint arXiv:1906.01820 (2019), https://arxiv.org/abs/1906.01820.

30 Kuiper, *Alistair Prime in a Box* (v1.0-preprint), Pillar 4 (Governance).

31 Kuiper, *Skipjack Protocol* (v1.0-preprint), §3 (Governance-Gated Execution).

32 Weslei Gomes de Sousa et al., "How and Where Is Artificial Intelligence in the Public Sector Going? A Literature Review and Research Agenda," *Government Information Quarterly* 36, no. 4 (2019): 101392, https://doi.org/10.1016/j.giq.2019.07.004.

33 Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger, "On Calibration of Modern Neural Networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML 2017)* 70 (2017): 1321–1330, https://proceedings.mlr.press/v70/guo17a.html.

miscalibrated, communicating false confidence in the very moments where the operator most needs accurate uncertainty information.

2.5 CONTEXT CURRENCY FAILURE

The fifth failure is **context currency failure**. Agentic systems assemble their operating context from multiple sources: prompt history, retrieved documents, tool outputs, persistent memory, latent assumptions derived from training. Each of these has its own currency horizon — some refresh every interaction, some persist across sessions, some are effectively immutable. Context currency failure is the condition in which the system treats sources of different currency as though they were uniformly fresh, with no architectural mechanism to discriminate between what has been validated *now* and what has been inherited from earlier state.

This problem is broader than dataset shift, although dataset shift is one of its instances.[35] Dataset shift treats the divergence between training distribution and deployment distribution; context currency failure treats the divergence between *currently authoritative* state and *currently used* state within a single deployment over time. NIST's lifecycle approach to AI risk management explicitly assumes that context cannot be settled once and treated as fixed; it must be managed.[36] The systems-level requirement is that the agentic system be able to mark, for each piece of state it is using, whether that piece has been refreshed within the current operating window or is being inherited from earlier state — and the structural failure is the absence of any architectural mechanism that makes this distinction available.

2.6 THE UNIFIED PROBLEM STATEMENT

Taken together, the five comprehension failures support a unified problem statement: **agentic systems fail when they cannot accurately represent, to themselves and to their operators, what they are doing, what they are using as authoritative context, and where they sit relative to the envelope they were authorized to operate within**. This is a structural deficiency, not a capability deficiency. It cannot be remedied by larger models, better prompts, or more training data alone. It requires the addition of an architectural component dedicated to the comprehension function — which is the subject of Section 3.

3. THE COMPREHENSION INSTRUMENT: FIVE FUNCTIONS

The **Comprehension Instrument** is proposed as a governance-first architectural component composed of five tightly coupled functions: **State Introspection (S)**, **Intent Rehearsal (I)**, **Envelope-Distance Estimation (D)**, **Drift Detection (T)**, and **Scope-Boundary Signaling (B)**. The order is deliberate. State Introspection anchors what the system knows about itself; Intent Rehearsal anchors what the system believes it was authorized to do; Envelope-Distance Estimation locates the system relative to its operational boundary; Drift Detection identifies movement of the operational position over time; Scope-Boundary Signaling exposes the relevant comprehension state to humans and to the orchestration layer for governance action.

34 Kuiper, *Skipjack Protocol* (v1.0-preprint), §3 (Explicit Uncertainty Signaling).

35 Joaquin Quionero-Candela, Masashi Sugiyama, Anton Schwaighofer, and Neil D. Lawrence, *Dataset Shift in Machine Learning* (Cambridge, MA: MIT Press, 2009).

36 NIST, *AI RMF 1.0*, lifecycle framing in §3.

The instrument is **not** a model. It is an architectural component whose functions can be implemented across a heterogeneous set of mechanisms — explicit programmatic checks, learned classifiers, structured prompts, runtime telemetry — but whose specification is invariant across implementation choices. The framework is generative rather than prescriptive in the same way that the HGC³AE² framework itself is: it specifies what must be present and how the parts must interact, leaving room for implementations that suit the substrate they run on.[37]

3.1 STATE INTROSPECTION (S)

State Introspection is the function by which the system produces, on demand and continuously, an accurate report of its current operating state. The function answers questions of the form: *What context am I currently treating as authoritative? Which of my working assumptions were established within the current session, and which are inherited? What is the currency horizon of each major state element I am using?* State Introspection is the system-internal analogue of the operator-facing transparency that NIST's measure function and human–AI interaction guidelines emphasize.[38][39]

In implementation, State Introspection requires that the system's runtime maintain a structured representation of its state with explicit currency metadata. This is not a logging concern; logs are post-hoc and operator-facing. State Introspection is real-time and self-facing: the agent's planning step must be able to query its own state representation as part of its decision logic, and the state representation must distinguish between freshly validated state and inherited state. The agentic substrate paper specifies a persistence plane that supports this requirement; State Introspection is the runtime use of that plane during execution.[40]

3.2 INTENT REHEARSAL (I)

Intent Rehearsal is the function by which the system periodically restates, in operationally meaningful terms, the intent it understands itself to be executing against. Intent Rehearsal is not a paraphrase task. It is a structured comparison between the system's current operative intent and the intent that the governance layer authorized, with the result exposed both to the system's own planning logic and to the orchestrator. When the rehearsed intent diverges from the authorized intent in a way the system can detect, the divergence is itself a governance signal.[41]

The cognitive-science analogue is the metacognitive monitoring function described by Nelson and Narens: the meta-level periodically compares the object-level's current operative state against the goal that was set, and the discrepancy is the signal that drives meta-level control.[42] In agentic terms, Intent Rehearsal is the architectural enforcement of this comparison. A system that does not rehearse its intent cannot detect the silent drift identified in §2.2; a

37 Kuiper, *HGC³AE² Framework* (v1.0-preprint), §3.

38 NIST, *AI RMF 1.0*, Measure function.

39 Saleema Amershi et al., "Guidelines for Human-AI Interaction," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (2019), https://doi.org/10.1145/3290605.3300233.

40 Kuiper, *Agentic Substrate* (v1.0-preprint), Persistence Plane.

41 Kuiper, *Operating Model* (v1.0-preprint), §2 (Governance Layer).

42 Thomas O. Nelson and Louis Narens, "Metamemory: A Theoretical Framework and New Findings," in *The Psychology of Learning and Motivation*, ed. Gordon H. Bower, vol. 26 (San Diego: Academic Press, 1990), 125–173, https://doi.org/10.1016/S0079-7421(08)60053-5.

system that rehearses but does not expose the rehearsal cannot trigger reconciliation checkpoints; a system that exposes the rehearsal but does not act on detected divergences has not fully integrated the function with the surrounding governance layer.

3.3 ENVELOPE-DISTANCE ESTIMATION (D)

Envelope-Distance Estimation is the function by which the system maintains a continuous estimate of where it currently sits relative to the boundaries of its pre-authorized decision envelope. The estimate need not be precise; what it must be is *available* — exposed as part of the system's runtime state in a way that the orchestrator can read and that the system's own planning logic can incorporate. Envelope-Distance Estimation transforms envelope blindness (§2.3) from a structural property of the system into a measurable runtime quantity.

In control-theoretic terms, Envelope-Distance Estimation is an observer for a particular class of state variables: the variables that describe the system's position in policy and authority space rather than in physical or computational state.[43] The observer need not be a Kalman filter; it can be implemented as a structured set of conditions that the system evaluates at each planning step, as a classifier that has been trained on examples of envelope-internal and envelope-external scenarios, or as a hybrid of both. What matters is that the function returns a usable estimate — a scalar distance, a categorical level, a structured set of warning flags — that downstream components can consume.

Public-sector AI literature has documented the consequences of operating systems without this function. de Sousa and colleagues' literature review of AI in the public sector finds repeatedly that the failures with the highest accountability cost are not failures of capability but failures of *bounding*: the system performs an action that no operator authorized, often because no one — neither the system nor its operators — could tell, in advance, that the action was outside the authorized envelope.[44] Envelope-Distance Estimation is the architectural response.

3.4 DRIFT DETECTION (T)

Drift Detection is the function by which the system identifies movement, over operational time, in the quantities that the other comprehension functions track. State introspection answers *what state am I in now?*; drift detection answers *how has my state changed since the last reconciliation point?*. Intent rehearsal answers *what intent am I executing?*; drift detection answers *how has my operative intent moved relative to the originally authorized intent?* Envelope-distance estimation answers *where do I sit relative to the boundary?*; drift detection answers *am I trending toward the boundary or away from it?*

The peer-reviewed literature on concept drift in machine-learning systems provides a substantial methodological foundation for this function. Gama and colleagues' survey of concept drift detection methods documents a range of statistical and learning-based techniques for identifying changes in the distributions over which a system operates.[45] In the agentic context,

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43 Rudolf E. Kalman, "A New Approach to Linear Filtering and Prediction Problems," *Journal of Basic Engineering* 82, no. 1 (1960): 35–45, https://doi.org/10.1115/1.3662552.

44 Weslei Gomes de Sousa et al., "How and Where Is Artificial Intelligence in the Public Sector Going? A Literature Review and Research Agenda," *Government Information Quarterly* 36, no. 4 (2019): 101392, https://doi.org/10.1016/j.giq.2019.07.004.

45 João Gama, Indrė Žliobaitė, Albert Bifet, Mykola Pechenizkiy, and Abdelhamid Bouchachia, "A Survey on Concept Drift Adaptation," *ACM Computing Surveys* 46, no. 4 (2014): article 44, https://doi.org/10.1145/2523813.

the relevant distributions include not only the data distribution that conventional concept-drift literature focuses on, but also the distributions over context-source currency, operative-intent specification, and envelope position. The instrument's drift detection function is the agentic generalization of the conventional concept-drift detector.

Crucially, Drift Detection in this framework is a metacognitive function: it operates over the outputs of the other comprehension functions, not directly over raw inputs. This is the architectural counterpart to Nelson and Narens' control-monitoring loop: the meta-level monitors the object-level's drift and can initiate meta-level control responses (recalibrating context, requesting human input, halting execution) on the basis of detected drift signals.[46]

3.5 SCOPE-BOUNDARY SIGNALING (B)

Scope-Boundary Signaling is the function by which the system communicates the relevant comprehension state to the orchestrator and to human operators. The function is the operational counterpart of Skipjack's *explicit uncertainty signaling* principle, generalized to cover not only confidence in outputs but also state currency, intent fidelity, envelope distance, and drift.[47]

Signaling is not optional reporting. It is the mechanism by which the Comprehension Instrument couples to the rest of the governance machinery. A system whose introspection, rehearsal, estimation, and drift detection all function but which does not signal results to the orchestrator is the same, operationally, as a system that lacks those functions altogether. The orchestrator's task routing decisions, the governance layer's envelope updates, and the operator's evaluation of system outputs all depend on signaling. Human–AI interaction guidelines have repeatedly emphasized this point: meaningful operator oversight requires that the system surface, in real time, the information operators need to calibrate reliance.[48][49]

3.6 THE MUTUALLY REINFORCING PROPERTY

A key strength of the framework is that the five functions are not modular conveniences; they are mutually reinforcing requirements. State Introspection without Intent Rehearsal yields a system that knows what it is doing but not whether it was supposed to be doing it. Intent Rehearsal without Envelope-Distance Estimation yields a system that compares intents but cannot place itself in the authority space. Envelope-Distance Estimation without Drift Detection yields a system that knows where it is but not where it is moving. Drift Detection without Scope-Boundary Signaling yields a system that detects motion but cannot communicate it to anyone who can act. Scope-Boundary Signaling without the other four functions yields a system that produces empty status reports. The instrument is generative precisely because the parts reinforce each other; it is not a checklist.

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46 Thomas O. Nelson and Louis Narens, "Metamemory: A Theoretical Framework and New Findings," in *The Psychology of Learning and Motivation*, ed. Gordon H. Bower, vol. 26 (San Diego: Academic Press, 1990), 125–173, https://doi.org/10.1016/S0079-7421(08)60053-5.

47 Kuiper, *Skipjack Protocol* (v1.0-preprint), §3 (Explicit Uncertainty Signaling).

48 Saleema Amershi et al., "Guidelines for Human-AI Interaction," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (2019), https://doi.org/10.1145/3290605.3300233.

49 Kevin Anthony Hoff and Masooda Bashir, "Trust in Automation: Integrating Empirical Evidence on Factors That Influence Trust," *Human Factors* 57, no. 3 (2015): 407–434, https://doi.org/10.1177/0018720814547570.

4. COMPREHENSION AS A CONTINUOUS OPERATIONAL PROCESS

A central claim of this paper is that **comprehension in agentic systems is not an event but a process**. The same temptation that affects the collaboration analysis in earlier work applies here: it is tempting to treat comprehension as something the system either has or lacks at the point of design and to evaluate the system once, statically.[50] For agentic systems, this view is inadequate. Comprehension state is itself dynamic. State changes; intent moves; envelopes are updated; drift accumulates. A snapshot of comprehension at deployment time does not predict comprehension state at operational time.

This temporal property aligns with the broader treatment of trustworthiness in NIST's AI RMF, which explicitly treats trustworthy AI as something governed and managed across the lifecycle rather than verified at a single point.[51][52] It also aligns with human–AI interaction research, which has long established that effective operator collaboration depends on the system's behavior over time, not only at initial interaction.[53]

The comprehension process can be described in three recurring phases: **maintenance**, **invocation**, and **reconciliation**.

4.1 MAINTENANCE

In the **maintenance** phase, the system continuously updates the state representations that the five comprehension functions read from. State Introspection's currency-marked state representation is refreshed as new context enters the system. Intent Rehearsal's record of authorized intent is updated when the governance layer revises the operating brief. Envelope-Distance Estimation's classifier or rule set is evaluated against new conditions. Drift Detection accumulates the time-series data over which its monitors operate. Scope-Boundary Signaling's output channels are kept active and addressable.

Maintenance is not glamorous, but it is load-bearing. A system whose comprehension state is stale will produce confident misalignment in proportion to the staleness, for the same reasons that any system operating on stale state produces incorrect outputs in proportion to the staleness. Tanenbaum and van Steen's treatment of distributed systems names this property explicitly: the freshness of state is a first-class concern in any system whose state is not single-process.[54] Kleppmann's treatment of data-intensive applications generalizes the point.[55] Comprehension state, in this framework, is no different from any other state in this regard.

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50 Kuiper, *HGC³AE² Framework* (v1.0-preprint), §4.

51 NIST, *AI RMF 1.0*, lifecycle framing in §3.

52 NIST, "Roadmap for the NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0)," accessed 2026-05-11, https://www.nist.gov/itl/ai-risk-management-framework/roadmap-nist-artificial-intelligence-risk-management-framework-ai.

53 Saleema Amershi et al., "Guidelines for Human-AI Interaction," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (2019), https://doi.org/10.1145/3290605.3300233.

54 Andrew S. Tanenbaum and Maarten van Steen, *Distributed Systems*, 3rd ed. (2017).

55 Martin Kleppmann, *Designing Data-Intensive Applications* (Sebastopol, CA: O'Reilly, 2017).

4.2 INVOCATION

In the **invocation** phase, the system's planning logic queries the comprehension functions as part of its decision-making. A planning step that is about to invoke a tool queries Envelope-Distance Estimation to determine whether the tool invocation is inside the authorized envelope. A planning step that is about to extend a context window queries State Introspection to determine the currency of the existing context. A planning step that is about to commit to a multi-turn trajectory queries Intent Rehearsal to confirm that the trajectory is still aligned with the authorized intent.

Invocation transforms comprehension from a property that the system reports about itself into a property that *gates* the system's actions. This is the architectural enforcement of metacognitive control as Nelson and Narens describe it: the meta-level's outputs do not merely report on the object-level, they shape what the object-level may do next.[56] In agentic terms, the comprehension functions are not telemetry — they are inputs to the planning loop.

4.3 RECONCILIATION

In the **reconciliation** phase, the system's accumulated comprehension state is compared against the authoritative state held by the governance layer and the operator, and any divergence is surfaced. Reconciliation is the agentic counterpart of the periodic re-establishment of shared situational awareness in high-reliability environments such as aviation and aerospace.[57][58] The Skipjack Protocol formalizes reconciliation checkpoints at meaningful transitions — between sessions, before consequential action, after new data integration, when authority conditions change.[59] The Comprehension Instrument provides the substantive content of those checkpoints: it is the introspection, rehearsal, estimation, and drift data that gets reconciled at each checkpoint.

Reconciliation that does not have substantive content to reconcile becomes ceremonial — a checkpoint that fires regularly but produces no actionable information. This is one of the failure modes the framework is meant to prevent. The five comprehension functions exist precisely so that reconciliation checkpoints have material to reconcile against.

4.4 THE CYCLE AS BIDIRECTIONAL ACCOUNTABILITY

The maintenance–invocation–reconciliation cycle is not a one-way reporting flow from the system to the operator. It is a bidirectional accountability mechanism. The human side of the boundary updates intent, constraints, and contextual corrections; the system side surfaces state, uncertainty, and dependencies. When either side stops doing so, shared situational awareness collapses — a pattern that has been repeatedly observed in human-systems

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56 Thomas O. Nelson and Louis Narens, "Metamemory: A Theoretical Framework and New Findings," in *The Psychology of Learning and Motivation*, ed. Gordon H. Bower, vol. 26 (San Diego: Academic Press, 1990), 125–173, https://doi.org/10.1016/S0079-7421(08)60053-5.

57 Federal Aviation Administration, *Human Factors Design Standard*, DOT/FAA/HF-STD-001B (Atlantic City International Airport, NJ: William J. Hughes Technical Center, 2016), https://hf.tc.faa.gov/hfds/.

58 National Aeronautics and Space Administration, *Human Integration Design Handbook (HIDH)*, NASA/SP-2010-3407 (Washington, DC: NASA, 2010), https://www.nasa.gov/reference/human-integration-design-handbook/.

59 Kuiper, *Skipjack Protocol* (v1.0-preprint), §3 (Reconciliation Checkpoints).

integration work in safety-critical aviation, aerospace, and command-and-control environments.[60][61] The Comprehension Instrument is the mechanism by which the system holds up its end of this reciprocity.

5. THE SKIPJACK–COMPREHENSION COUPLING

The Skipjack Protocol's four enforcement mechanisms are the runtime layer at which the Comprehension Instrument's outputs do their work. This section walks each mechanism and identifies where the comprehension functions feed it.

5.1 MECHANISM 1 — CONTEXT INTEGRITY AND EPISTEMIC CONTINUITY

Skipjack's first mechanism enforces that the context the system operates on is integral — refreshed when it must be, validated when it can be, and discriminated from stale state when refresh is not possible.[62] The Comprehension Instrument feeds this mechanism through State Introspection and through Drift Detection over State Introspection's outputs. A system that cannot describe its own context state cannot enforce integrity over that state. A system that cannot detect drift in its context state cannot identify the moments at which integrity has been violated.

The pathological case is the system that *reports* clean context state because it has never refreshed the currency metadata that would expose stale state. The Comprehension Instrument addresses this by making currency metadata a first-class state element, not a derived one; State Introspection cannot return clean state if the metadata has not been refreshed within the current operating window.

5.2 MECHANISM 2 — STRUCTURED TASK DECOMPOSITION AND ROUTING

Skipjack's second mechanism enforces that tasks are decomposed and routed within authorized boundaries.[63] The Comprehension Instrument feeds this mechanism through Envelope-Distance Estimation and through Scope-Boundary Signaling. The orchestrator's routing logic — formalized in the operating-model paper as the orchestrator function in the Scrum-to-Skipjack remap[64] — consumes envelope-distance estimates and scope-boundary signals to determine whether a proposed routing is safe, whether it needs to be escalated to the governance layer, or whether it should be halted entirely.

A system that has accurate envelope-distance estimates but does not signal them to the orchestrator effectively has no envelope. The orchestrator routes based on what it knows; what it knows is what the system signals. The coupling between Envelope-Distance Estimation and

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60 National Aeronautics and Space Administration, *Human Integration Design Handbook (HIDH)*, NASA/SP-2010-3407 (Washington, DC: NASA, 2010), https://www.nasa.gov/reference/human-integration-design-handbook/.

61 Federal Aviation Administration, *Human Factors Design Standard*, DOT/FAA/HF-STD-001B (Atlantic City International Airport, NJ: William J. Hughes Technical Center, 2016), https://hf.tc.faa.gov/hfds/.

62 Kuiper, *Skipjack Protocol* (v1.0-preprint), Mechanism 1.

63 Kuiper, *Skipjack Protocol* (v1.0-preprint), Mechanism 2.

64 Justin H. Kuiper, CISSP, *The Operating Model for Agentic Teams* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869313.

Scope-Boundary Signaling is therefore not optional; it is the gating mechanism by which envelope authority becomes operational.

5.3 MECHANISM 3 — HUMAN-IN-THE-LOOP CONTROL POINTS

Skipjack's third mechanism is the operational instantiation of human-in-the-loop authority — discrete points at which a human operator must validate or authorize a continuation.[65] The Comprehension Instrument feeds this mechanism through Intent Rehearsal and through Scope-Boundary Signaling. An HITL control point that fires when intent rehearsal detects divergence is operationally meaningful; an HITL control point that fires on a fixed schedule regardless of system state is ceremonial.

The general principle is that HITL control points should fire when comprehension signals indicate that human judgment is required, and should not fire when comprehension signals indicate that the system is operating well inside its envelope with stable comprehension state. This both reduces ceremonial overhead and increases the substantive content of each control point. Operator attention is finite; the Comprehension Instrument is what allows it to be deployed at moments of actual decision rather than moments of arbitrary scheduling.[66]

5.4 MECHANISM 4 — ITERATIVE FEEDBACK LOOPS

Skipjack's fourth mechanism is the feedback loop that closes between human judgment and agentic execution, allowing the system's operation to improve as the team accumulates experience.[67] The Comprehension Instrument feeds this mechanism through all five of its functions taken together. The feedback loop's substantive content — what the system learned in the last sprint that should change how it operates in the next sprint — is precisely the introspection, rehearsal, estimation, drift, and signaling data accumulated over the prior operating window.

Feedback loops without comprehension data degrade into either of two failure modes: empty retrospection ("the sprint went fine"), or unstructured complaint ("the system did things I didn't want"). With comprehension data, the feedback loop becomes a structured analysis of where the system's state, intent, envelope position, and drift differed from what the operator authorized and why. This is the loop closure that the HGC^3AE^2 framework's exponent — the squared term that captures the closure of governance, context, and execution back into itself — depends on.[68]

6. IMPLICATIONS AND CONCLUSION

The implications of the Comprehension Instrument framework extend beyond AI system design into organizational governance, public administration, defense operations, and human systems integration. Public-sector AI research has already shown that AI deployment affects policy cycles, citizen interaction, and institutional accountability in ways that depend critically on

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65 Kuiper, *Skipjack Protocol* (v1.0-preprint), Mechanism 3.

66 Kevin Anthony Hoff and Masooda Bashir, "Trust in Automation: Integrating Empirical Evidence on Factors That Influence Trust," *Human Factors* 57, no. 3 (2015): 407–434, https://doi.org/10.1177/0018720814547570.

67 Kuiper, *Skipjack Protocol* (v1.0-preprint), Mechanism 4.

68 Kuiper, *HGC^3AE^2 Framework* (v1.0-preprint), §3.

the operator's capacity to maintain meaningful oversight.[69][70] In defense settings, the stakes are higher still: the U.S. Department of Defense's public AI ethics principles — responsible, equitable, traceable, reliable, governable — are not satisfiable by a system whose own state, intent, and envelope position are opaque to its operators.[71] The principles' operational meaning depends on the existence of architectural mechanisms that make the system *governable* in real time, which is precisely what the Comprehension Instrument specifies.

The core claim of this paper is straightforward: **agentic systems fail not because they lack comprehension capability, but because they lack the architectural means of making their comprehension available to the governance machinery surrounding them**. Capability without legibility invites confident misalignment. Legibility without continuity yields stale comprehension. Continuity without coupling to Skipjack mechanisms produces telemetry that no one acts on. The HGC^{3}AE^2 framework's response — Human-governed, Curated Context, Agentically Engineered and Executed alignment, with the exponent capturing closure — is the architecture within which the Comprehension Instrument operates. The instrument is the runtime cognition that lets the framework's structural commitments hold under operational conditions.

Returned to the framework letter by letter:

The **Human-governed (H)** element supplies the authority that defines the envelope. The Comprehension Instrument does not redefine the envelope. Calibration parameters, drift thresholds, signaling channels, reconciliation cadence — all are governance-layer decisions, owned by humans accountable for the system's operation. An instrument whose own parameters are self-tuned by the agent has surrendered the H of HGC^{3}AE^2.

The **Curated Context (G/C)** element supplies the source material for State Introspection. The instrument does not synthesize its own context; it reads from the curated context tier specified in the agentic substrate paper.[72] State Introspection's currency metadata is the runtime expression of curation discipline.

The **Cybersecurity (C^1)** dimension is a load-bearing concern for the instrument itself. The comprehension functions are a privileged surface; tampering with them is a security event with consequences for every downstream governance mechanism. The security and provenance treatment in the Managing Agentics Ops series develops this surface in depth.[73]

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69 Weslei Gomes de Sousa et al., "How and Where Is Artificial Intelligence in the Public Sector Going? A Literature Review and Research Agenda," *Government Information Quarterly* 36, no. 4 (2019): 101392, https://doi.org/10.1016/j.giq.2019.07.004.

70 Gianluca Misuraca and Colin van Noordt, "Assessing the Public Policy-Cycle Framework in the Age of Artificial Intelligence," *Government Information Quarterly* 37, no. 4 (2020): 101509, https://doi.org/10.1016/j.giq.2020.101509.

71 U.S. Department of Defense, "DoD Adopts Ethical Principles for Artificial Intelligence," February 24, 2020, https://www.defense.gov/News/Releases/Release/Article/2091996/dod-adopts-ethical-principles-for-artificial-intelligence/.

72 Justin H. Kuiper, CISSP, *The Agentic Substrate* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869291.

73 Forthcoming MAO-P9 *Security & Provenance for Self-Hosted Agentic Systems* (v0.1-seed, in concurrent drafting; yks-build#102).

The **Continuity (C^2)** dimension is what Section 4 names: comprehension is a process, not an event, and the persistence plane of the agentic substrate is what enables the process to span sessions.[74]

The **Coherence (C^3)** dimension is what Drift Detection guards. Movement is not bad in itself; movement that drifts the system away from coherent operation is. The drift function specifies which movements matter and which can be ignored.

The **Agentically Engineered (A)** dimension is the implementation. The instrument is built by the system's runtime, exposed by its substrate, and operated by the agent — not bolted on as external monitoring. External monitoring sees what the agent surfaces; comprehension is part of what the agent surfaces.

The **Executed (E)** dimension is the invocation phase of Section 4. Comprehension functions execute continuously at runtime, not only at decision points.

The exponent — the closure — is the maintenance–invocation–reconciliation cycle. The instrument closes its own loop.

This paper has deliberately limited its scope to the architectural specification of the Comprehension Instrument and its coupling to the surrounding Mission-Ready framework. Implementation patterns across heterogeneous substrates — Omen on-prem nodes, hyperscaler-managed inference surfaces, sovereign bare-metal deployments — are the subject of the operational-plan paper in the same series.[75] Calibration procedures, parameter governance, and adversarial robustness of the comprehension functions themselves are subjects that warrant dedicated future treatment, and the Intent Horizon series is positioned to address several of them.[76] The present paper is sufficient to make the structural claim: a mission-ready agentic system without a functioning Comprehension Instrument cannot be said to operate within the HGC^3AE^2 governance frame, regardless of its capability or training.

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— end v0.1-draft —

Drafting state: v0.1 exemplar (chain v0.2 step 3). Shape-review checkpoint per exemplar-first cadence ratified 2026-05-11. On approval: roll same MR-P1 shape through remaining 20 round-1 papers.

Citation count this draft: 18 peer-reviewed sources cited inline (Flavell · Nelson & Narens · Schraw & Moshman · Premack & Woodruff · Rabinowitz et al. ICML 2018 · Kalman · Hubinger et al. arXiv · Russell · Guo et al. ICML 2017 · Gama et al. ACM Computing Surveys · Quionero-Candela et al. MIT Press · Arrieta et al. *Information Fusion* · Amershi et al. CHI 2019 · Hoff & Bashir *Human Factors* · de Sousa et al. *GIQ* · Misuraca & van Noordt *GIQ* · Tanenbaum & van

74 Justin H. Kuiper, CISSP, *The Agentic Substrate* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869291.

75 Forthcoming MR-P9 *Operational Plan — Heterogeneous Robo Stack Proof* (v1.0-preprint, in concurrent drafting; yks-build#99).

76 Intent Horizon Series, gate-blocked behind MR G3 + MAO G2 (yks-build#111–#120).

Steen · Kleppmann). Plus primary-source standards (NIST AI RMF, DoD ethics, NASA HIDH, FAA HFDS) and YKS canon inheritance (MR-P1, P3, P4, P5, P6, P7).

Page-floor preflight: v0.1 word count ~7800 → ≈ 22–24 typeset pages with v1.0-preprint formatting; expansion to confirmed ≥24pp at v0.5 with §3.x function depth, §5 mechanism walks expanded with example traces, §6 implementation-pattern discussion.